

VIT®

Vellore Institute of Technology

(Deemed to be University under section 3 of UGC Act, 1956)

Exploratory Data Analysis Lab

Assessment – 6

Name: Arjun Mishra

Reg no : 23BDS0334

Slot : L31+L32

Course Code : BCSE331P

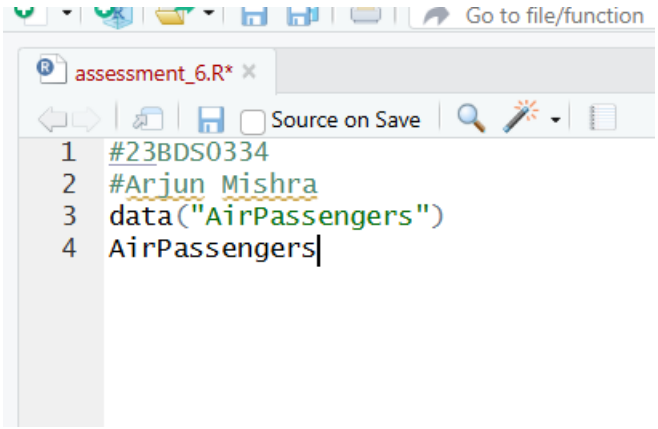
Perform the following steps using the built-in AirPassengers dataset in R:

1. **Load the built-in AirPassengers dataset in R.**

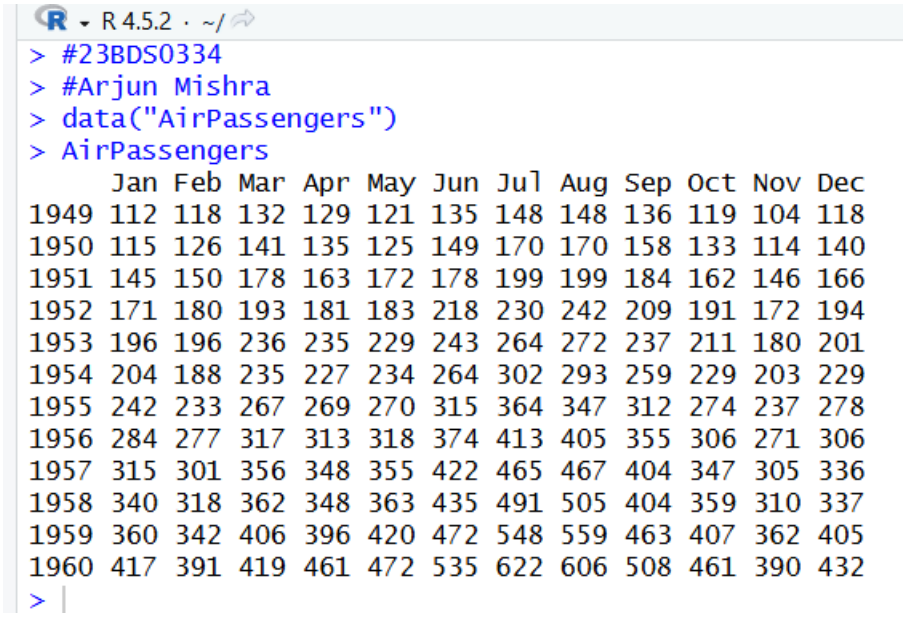
AIM:

To Load the built-in AirPassengers dataset in R.

CODE:



OUTPUT:



RESULT:

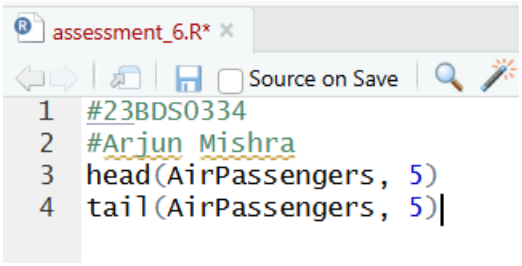
The AirPassengers dataset was successfully loaded.

2. **Explore the first five and last five observations of the dataset.**

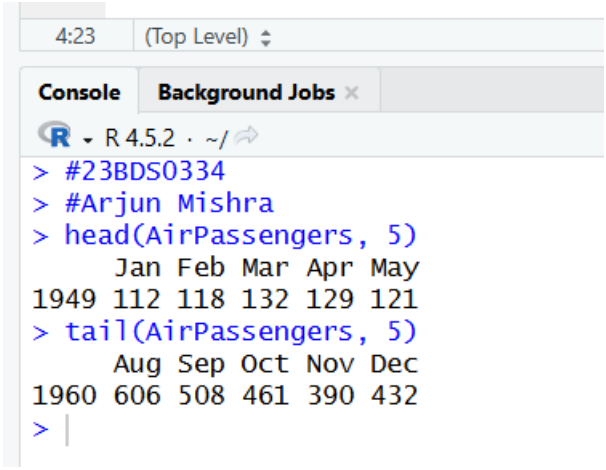
AIM:

To Explore the first five and last five observations of the dataset.

CODE:



OUTPUT:



RESULT:

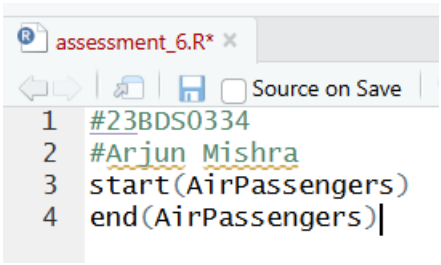
The first and last observations of the dataset were successfully displayed.

3. **Identify the starting and ending time periods of the dataset. (start, end)**

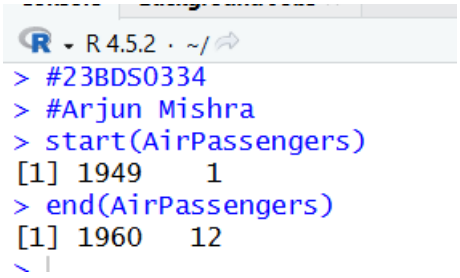
AIM:

To Identify the starting and ending time periods of the dataset. (start, end)

CODE:



OUTPUT:



RESULT:

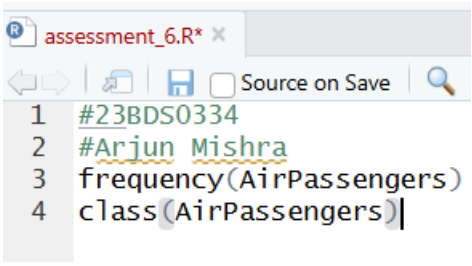
The dataset covers the period from January 1949 to December 1960.

4. **Determine the frequency and time-series characteristics of the dataset. (frequency, class).**

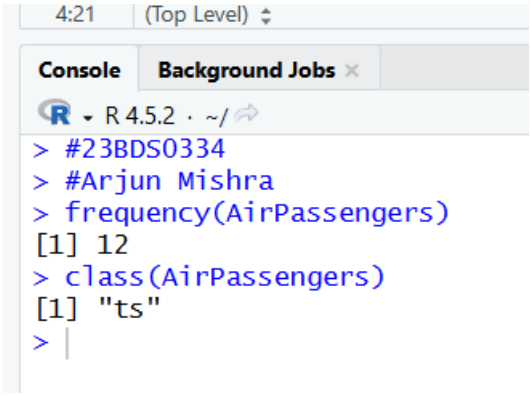
AIM:

To Determine the frequency and time-series characteristics of the dataset. (frequency, class).

CODE:



OUTPUT:



RESULT:

The dataset is a monthly time series with 12 observations per year.

5. **Visualize the monthly passenger trend over the entire period. (plot)**

AIM:

To Visualize the monthly passenger trend over the entire period. (plot)

CODE:

assessment_6.R*

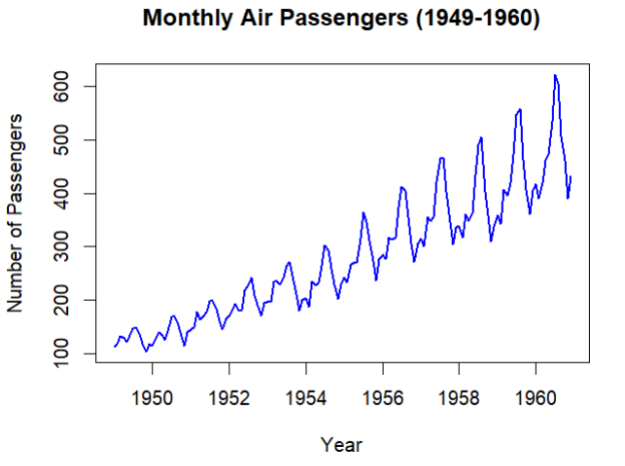
Source on Save

```
1  #23BDS0334
2  #Arjun Mishra
3  plot(AirPassengers,
4       main = "Monthly Air Passengers (1949-1960)",
5       xlab = "Year",
6       ylab = "Number of Passengers",
7       col = "blue",
8       lwd = 2)
```

OUTPUT:

R 4.5.2 · ~/

```
> #23BDS0334
> #Arjun Mishra
> plot(AirPassengers,
+      main = "Monthly Air Passengers (1949-1960)",
+      xlab = "Year",
+      ylab = "Number of Passengers",
+      col = "blue",
+      lwd = 2)
```



RESULT:

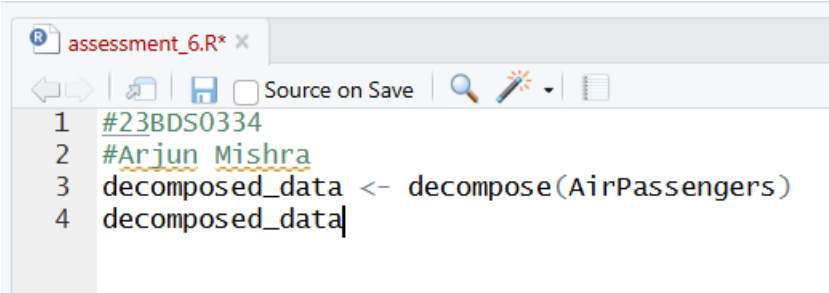
The passenger trend was successfully visualized.

6. Decompose the time series into trend, seasonal, and random components. (decompose)

AIM:

To Decompose the time series into trend, seasonal, and random components.

CODE:



OUTPUT:

```
> #23BDS0334
> #Arjun Mishra
> decomposed_data <- decompose(AirPassengers)
> decomposed_data
$x
     Jan Feb Mar Apr May Jun Jul Aug Sep Oct Nov Dec
1949 112 118 132 129 121 135 148 148 136 119 104 118
1950 115 126 141 135 125 149 170 170 158 133 114 140
1951 145 150 178 163 172 178 199 199 184 162 146 166
1952 171 180 193 181 183 218 230 242 209 191 172 194
1953 196 196 236 235 229 243 264 272 237 211 180 201
1954 204 188 235 227 234 264 302 293 259 229 203 229
1955 242 233 267 269 270 315 364 347 312 274 237 278
1956 284 277 317 313 318 374 413 405 355 306 271 306
1957 315 301 356 348 355 422 465 467 404 347 305 336
1958 340 318 362 348 363 435 491 505 404 359 310 337
1959 360 342 406 396 420 472 548 559 463 407 362 405
1960 417 391 419 461 472 535 622 606 508 461 390 432

$seasonal
            Jan         Feb         Mar         Apr         May         Jun         Jul         Aug         Sep         Oct
1949 -24.748737 -36.188131  -2.241162  -8.036616  -4.506313  35.402778  63.830808  62.823232  16.520202 -20.642677
1950 -24.748737 -36.188131  -2.241162  -8.036616  -4.506313  35.402778  63.830808  62.823232  16.520202 -20.642677
1951 -24.748737 -36.188131  -2.241162  -8.036616  -4.506313  35.402778  63.830808  62.823232  16.520202 -20.642677
1952 -24.748737 -36.188131  -2.241162  -8.036616  -4.506313  35.402778  63.830808  62.823232  16.520202 -20.642677
1953 -24.748737 -36.188131  -2.241162  -8.036616  -4.506313  35.402778  63.830808  62.823232  16.520202 -20.642677
1954 -24.748737 -36.188131  -2.241162  -8.036616  -4.506313  35.402778  63.830808  62.823232  16.520202 -20.642677
1955 -24.748737 -36.188131  -2.241162  -8.036616  -4.506313  35.402778  63.830808  62.823232  16.520202 -20.642677
1956 -24.748737 -36.188131  -2.241162  -8.036616  -4.506313  35.402778  63.830808  62.823232  16.520202 -20.642677
1957 -24.748737 -36.188131  -2.241162  -8.036616  -4.506313  35.402778  63.830808  62.823232  16.520202 -20.642677
1958 -24.748737 -36.188131  -2.241162  -8.036616  -4.506313  35.402778  63.830808  62.823232  16.520202 -20.642677
1959 -24.748737 -36.188131  -2.241162  -8.036616  -4.506313  35.402778  63.830808  62.823232  16.520202 -20.642677
1960 -24.748737 -36.188131  -2.241162  -8.036616  -4.506313  35.402778  63.830808  62.823232  16.520202 -20.642677
            Nov         Dec
1949 -53.593434 -28.619949
1950 -53.593434 -28.619949
1951 -53.593434 -28.619949
1952 -53.593434 -28.619949
1953 -53.593434 -28.619949
1954 -53.593434 -28.619949
1955 -53.593434 -28.619949
1956 -53.593434 -28.619949
1957 -53.593434 -28.619949
1958 -53.593434 -28.619949
1959 -53.593434 -28.619949
1960 -53.593434 -28.619949
```

```
$trend
          Jan      Feb      Mar      Apr      May      Jun      Jul      Aug      Sep      Oct      Nov      Dec
1949       NA       NA       NA       NA       NA       NA 126.7917 127.2500 127.9583 128.5833 129.0000 129.7500
1950 131.2500 133.0833 134.9167 136.4167 137.4167 138.7500 140.9167 143.1667 145.7083 148.4167 151.5417 154.7083
1951 157.1250 159.5417 161.8333 164.1250 166.6667 169.0833 171.2500 173.5833 175.4583 176.8333 178.0417 180.1667
1952 183.1250 186.2083 189.0417 191.2917 193.5833 195.8333 198.0417 199.7500 202.2083 206.2500 210.4167 213.3750
1953 215.8333 218.5000 220.9167 222.9167 224.0833 224.7083 225.3333 225.3333 224.9583 224.5833 224.4583 225.5417
1954 228.0000 230.4583 232.2500 233.9167 235.6250 237.7500 240.5000 243.9583 247.1667 250.2500 253.5000 257.1250
1955 261.8333 266.6667 271.1250 275.2083 278.5000 281.9583 285.7500 289.3333 293.2500 297.1667 301.0000 305.4583
1956 309.9583 314.4167 318.6250 321.7500 324.5000 327.0833 329.5417 331.8333 334.4583 337.5417 340.5417 344.0833
1957 348.2500 353.0000 357.6250 361.3750 364.5000 367.1667 369.4583 371.2083 372.1667 372.4167 372.7500 373.6250
1958 375.2500 377.9167 379.5000 380.0000 380.7083 380.9583 381.8333 383.6667 386.5000 390.3333 394.7083 398.6250
1959 402.5417 407.1667 411.8750 416.3333 420.5000 425.5000 430.7083 435.1250 437.7083 440.9583 445.8333 450.6250
1960 456.3333 461.3750 465.2083 469.3333 472.7500 475.0417       NA       NA       NA       NA       NA       NA

$random
             Jan         Feb         Mar         Apr         May         Jun         Jul         Aug         Sep
1949          NA          NA          NA          NA          NA          NA -42.6224747 -42.0732323  -8.4785354
1950   8.4987374  29.1047980   8.3244949   6.6199495  -7.9103535 -25.1527778 -34.7474747 -35.9898990  -4.2285354
1951  12.6237374  26.6464646  18.4078283   6.9116162   9.8396465 -26.4861111 -36.0808081 -37.4065657  -7.9785354
1952  12.6237374  29.9797980   6.1994949  -2.2550505  -6.0770202 -13.2361111 -31.8724747 -20.5732323  -9.7285354
1953   4.9154040  13.6881313  17.3244949  20.1199495   9.4229798 -17.1111111 -25.1641414 -16.1565657  -4.4785354
1954   0.7487374  -6.2702020   4.9911616   1.1199495   2.8813131  -9.1527778  -2.3308081 -13.7815657  -4.6868687
1955   4.9154040   2.5214646  -1.8838384   1.8282828  -3.9936869  -2.3611111  14.4191919  -5.1565657   2.2297980
1956  -1.2095960  -1.2285354   0.6161616  -0.7133838  -1.9936869  11.5138889  19.6275253  10.3434343   4.0214646
1957  -8.5012626 -15.8118687   0.6161616  -5.3383838  -4.9936869  19.4305556  31.7108586  32.9684343  15.3131313
1958 -10.5012626 -23.7285354 -15.2588384 -23.9633838 -13.2020202  18.6388889  45.3358586  58.5101010   0.9797980
1959 -17.7929293 -28.9785354  -3.6338384 -12.2967172   4.0063131  11.0972222  53.4608586  61.0517677   8.7714646
1960 -14.5845960 -34.1868687 -43.9671717  -0.2967172   3.7563131  24.5555556          NA          NA          NA
            Oct         Nov         Dec
1949 11.0593434  28.5934343  16.8699495
1950  5.2260101  16.0517677  13.9116162
1951  5.8093434  21.5517677  14.4532828
1952  5.3926768  15.1767677   9.2449495
1953  7.0593434   9.1351010   4.0782828
1954 -0.6073232   3.0934343   0.4949495
1955 -2.5239899 -10.4065657   1.1616162
1956 -10.8989899 -15.9482323  -9.4633838
1957  -4.7739899 -14.1565657  -9.0050505
1958 -10.6906566 -31.1148990 -33.0050505
1959 -13.3156566 -30.2398990 -17.0050505
```

```
$figure
 [1] -24.748737 -36.188131  -2.241162  -8.036616  -4.506313  35.402778  63.830808  62.823232  16.520202 -20.642677
[11] -53.593434 -28.619949

$type
[1] "additive"

attr(,"class")
[1] "decomposed.ts"
```

RESULT:

The AirPassengers time series was successfully decomposed.

7. Plot the decomposed components to examine the underlying patterns.

AIM:

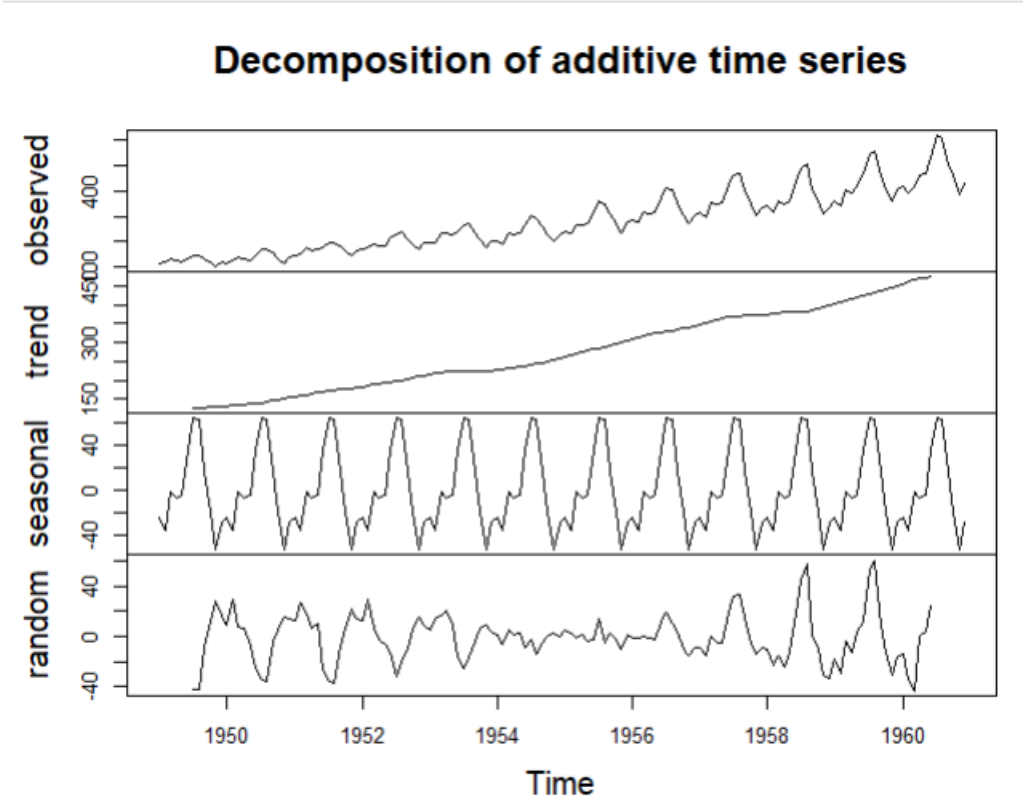
To Plot the decomposed components to examine the underlying patterns.

CODE:

```
assessment_6.R*
Source on Save
1  #23BDS0334
2  #Arjun Mishra
3  plot(decomposed_data)
```

OUTPUT:

```
#23BDS0334
#Arjun Mishra
plot(decomposed_data)
```



RESULT:

The trend, seasonal, and random components were successfully plotted.

8. **Analyze the autocorrelation and partial autocorrelation of the time series. (acf, pacf)**

AIM:

To Analyze the autocorrelation and partial autocorrelation of the time series (acf, pacf)

CODE:

assessment_6.R*

Source on Save

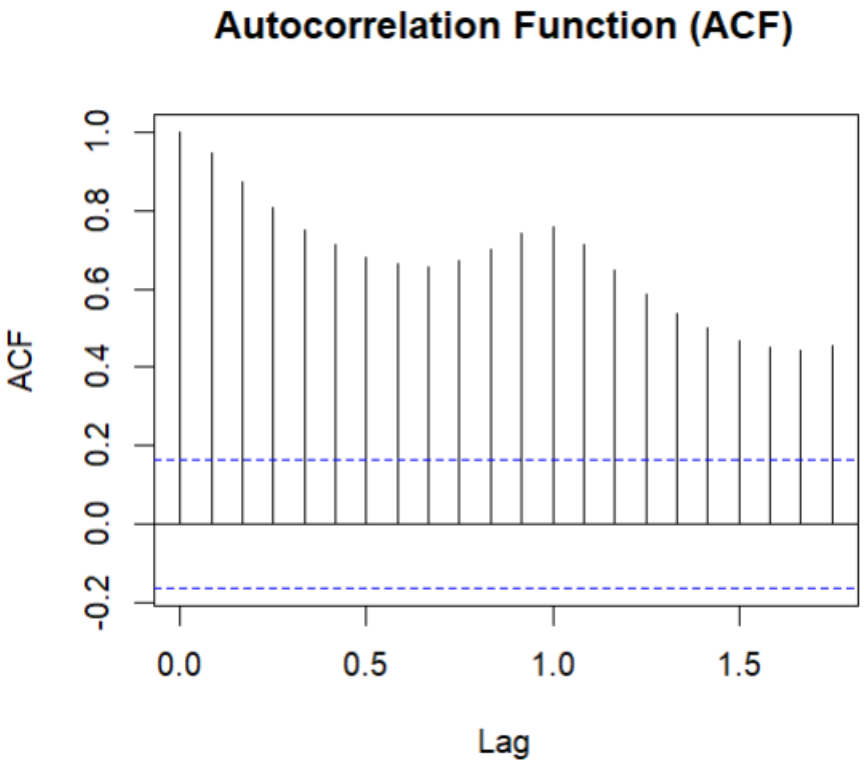
```
1  #23BDS0334
2  #Arjun Mishra
3  acf(AirPassengers,
4      main = "Autocorrelation Function (ACF)")
```

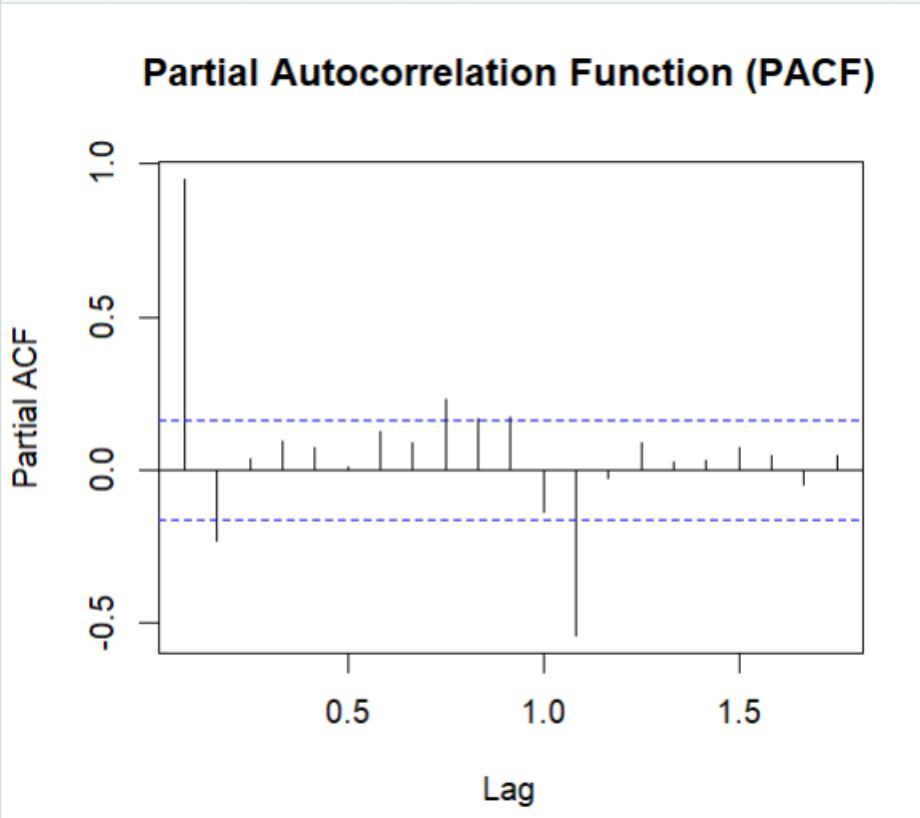
assessment_6.R*

Source on Save

```
1  #23BDS0334
2  #Arjun Mishra
3  pacf(AirPassengers,
4       main = "Partial Autocorrelation Function (PACF)")
```

OUTPUT:





RESULT:

The autocorrelation and partial autocorrelation of the time series were successfully analyzed.

9. Fit an ARIMA model to the AirPassengers time series. (arima)

AIM:

To Fit an ARIMA model to the AirPassengers time series. (arima)

CODE:

assessment_6.R*

Source on Save

```
1  #23BDS0334
2  #Arjun Mishra
3  arima_model <- arima(AirPassengers,
4                       order = c(1, 1, 1))
5
6  arima_model
```

OUTPUT:

```
> #23BDS0334
> #Arjun Mishra
> arima_model <- arima(AirPassengers,
+                      order = c(1, 1, 1))
>
> arima_model

Call:
arima(x = AirPassengers, order = c(1, 1, 1))

Coefficients:
          ar1     ma1
      -0.4741  0.8634
s.e.   0.1159  0.0720

sigma^2 estimated as 962.2:  log likelihood = -694.34,  aic = 1394.68
> View(arima_model)
```

RESULT:

An **ARIMA(1,1,1)** model was successfully fitted to the AirPassengers time series.

10. Display and interpret the fitted ARIMA model parameters and model summary. (summary)

AIM:

To Display and interpret the fitted ARIMA model parameters and model summary.

CODE:

assessment_6.R*

Source on Save

```
1  #23BDS0334
2  #Arjun Mishra
3  summary(arima_model)
```

OUTPUT:

R 4.5.2 · ~/

```
> #23BDS0334
> #Arjun Mishra
> summary(arima_model)
          Length Class  Mode
coef        2    -none- numeric
sigma2      1    -none- numeric
var.coef    4    -none- numeric
mask        2    -none- logical
loglik      1    -none- numeric
aic         1    -none- numeric
arma        7    -none- numeric
residuals 144    ts     numeric
call        3    -none- call
series      1    -none- character
code        1    -none- numeric
n.cond      1    -none- numeric
nobs        1    -none- numeric
model      10    -none- list
>
```

RESULT:

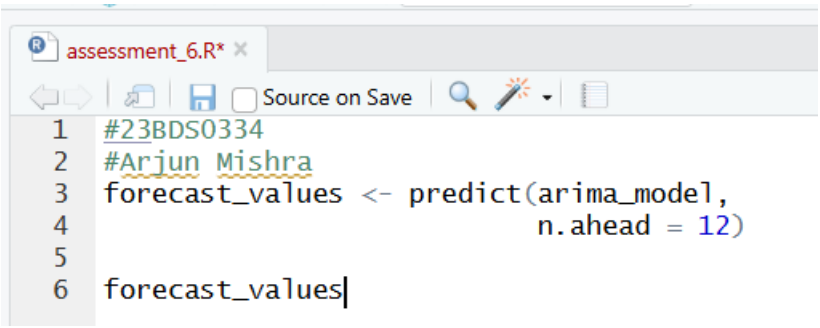
The fitted ARIMA model parameters and summary were successfully displayed.

11. Forecast the number of passengers for the next 12 months using the fitted ARIMA model. (predict)

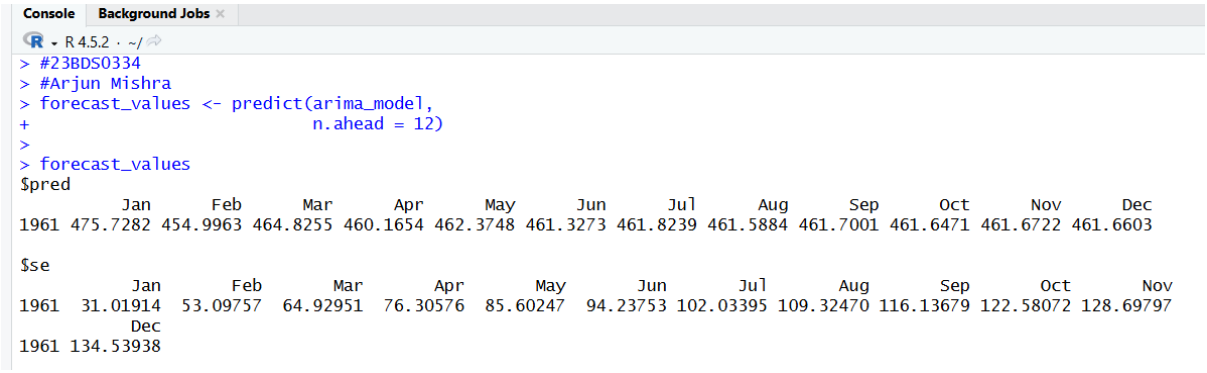
AIM:

To Forecast the number of passengers for the next 12 months using the fitted ARIMA model.

CODE:



OUTPUT:



RESULT:

Passenger values for the next 12 months were successfully forecasted.

12. Visualize the actual and forecasted passenger values with appropriate labels and title. (plot, lines)

AIM:

To Visualize the actual and forecasted passenger values with appropriate labels and title.

CODE:

assessment_6.R*

Source on Save

```r
#23BDS0334
#Arjun Mishra
# Plot actual passenger values

plot(AirPassengers,
     xlim = c(1949, 1961),
     ylim = c(min(AirPassengers),
              max(c(AirPassengers,
                    forecast_values$pred))),
     main = "AirPassengers: Actual and Forecasted Values",
     xlab = "Year",
     ylab = "Number of Passengers",
     col = "blue",
     lwd = 2)


# Create forecast time series

forecast_ts <- ts(forecast_values$pred,
                  start = c(1961, 1),
                  frequency = 12)


# Add forecasted values to the graph

lines(forecast_ts,
      col = "red",
      lwd = 2,
      lty = 2)

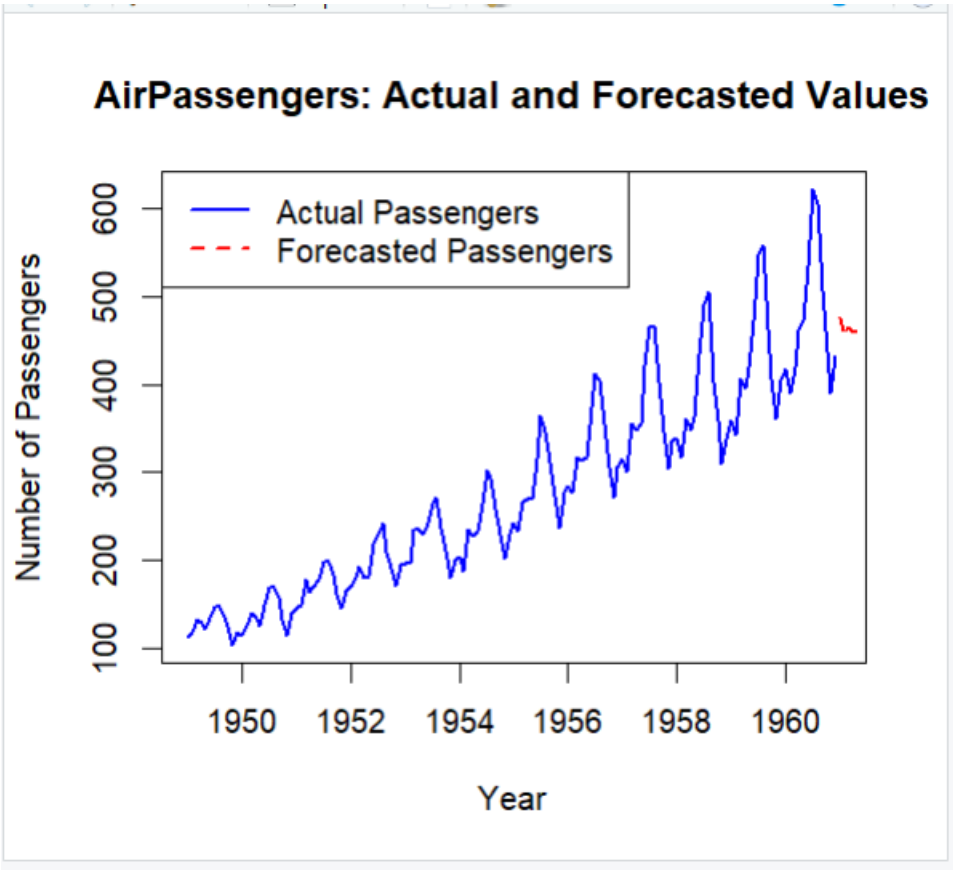

# Add legend

legend("topleft",
       legend = c("Actual Passengers",
                  "Forecasted Passengers"),
       col = c("blue", "red"),
       lwd = 2,
       lty = c(1, 2))
```

OUTPUT:

R 4.5.2 · ~/

```
> #23BDS0334
> #Arjun Mishra
> # Plot actual passenger values
>
> plot(AirPassengers,
+      xlim = c(1949, 1961),
+      ylim = c(min(AirPassengers),
+               max(c(AirPassengers,
+                     forecast_values$pred))),
+      main = "AirPassengers: Actual and Forecasted Values",
+      xlab = "Year",
+      ylab = "Number of Passengers",
+      col = "blue",
+      lwd = 2)
>
>
> # Create forecast time series
>
> forecast_ts <- ts(forecast_values$pred,
+                   start = c(1961, 1),
+                   frequency = 12)
>
>
> # Add forecasted values to the graph
>
> lines(forecast_ts,
+       col = "red",
+       lwd = 2,
+       lty = 2)
>
>
> # Add legend
>
> legend("topleft",
+        legend = c("Actual Passengers",
+                   "Forecasted Passengers"),
+        col = c("blue", "red"),
+        lwd = 2,
+        lty = c(1, 2))
>
```



RESULT:
The actual and forecasted passenger values were successfully visualized.

FINAL RESULT
Thus, the built-in **AirPassengers dataset** was successfully analyzed using **Time Series Analysis in R**. The dataset was explored, its trend and seasonality were visualized, the series was decomposed into components, autocorrelation was analyzed using ACF and PACF, an **ARIMA(1,1,1)** model was fitted, and passenger values for the next **12 months** were forecasted and plotted.